SQL Aggregate and operations

Exercise 2

1. Select distinct (department)
   from Students;

department
IT
HR
Finance
*

2. Select department, Avg (age) as Avg_age
   from Students

Department	Avg_age
IT	20.5
HR	22
Finance	23

3. Select department, Count (*) As Student_Co
   from Students
   Group by department
   Having Count (*) > 1;

department	Student_Count
IT	2
HR	2

4. Select student_id, name, age, department
from students
where age between 21 and 23;

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Dana	23	finance

5. Select student_id, name, age, department
from students
where departments in ('IT', 'HR')
and age > 21;

students_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. Select department, sum(credits) as total_credits
from courses
Group by department
Having sum(credits) > 5;

department	total_credits
IT	11

7. Select Course_Id, Course_Name, department, Credits
from Courses
where Credits <> 4

Course_Id	Course_Name	department	Credits
101	SQL basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. select Course_Id, Course_Name, Credits
from Courses
order by Credits Desc
Limit 3;

Course_Id	Course_Name	Credits
102	Python	4
103	Data Science	4
101	SQL basics	3.

9. select Max (grade) as Max_grade
min (grade) as min_grade
AVG (grade as Avg_grade
from enrollments;

Max_grad	min_grade	Avg_grade
90	78	84.6

10. select course_id, count (*) as enrollment_co
from enrollments
Group by course_id;

Course_id	enrollment_Count
101	1
102	1
103	1
104	1
105	1

11. select department
sum (salary) as total_salary
sum (bonus) as total_bonus
from salaries
Group by department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
finance	70000	6800

12. select department, avg (salary) as AVG salary
from salaries
Group by department
Having AVG (salary) > 55000

department	Avg_salary
IT	61000
finance	70000

13. Select employee_id, name, salary, bonus, (salary + bonus) as Total_Compensation
from salaries
where (salary + bonus) > 60000;

employee_id	name	salary	bonus	total_Compensation
1	TOM	60000	5000	65000
3	spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. Select department, Sum (budget) as total_budget
~~AVG~~ (budget) as Avg_budget
from [redacted] projects.
Group by department.
Having AVG (budget) > 70000;

department	total budget	Avg_budget
IT	270 000	135 000
finance.	85000	85000

15. select project_id, project_name, department, budget
from projects
where budget between 50000 and 120000
and department <> 'Marketing';

project_id	project_name	department	budget
1	AI.APP	IT	120000
2	payroll system	finance	80000
4	website	marketing	60000
5	HR portal	HR	50000